A Multilingual Voice-Enabled Dual-Index RAG Framework for Agricultural Advisory and Government Scheme Assistance

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***Abstract*—**Smallholder farmers need two kinds of information that have very little in common. One concerns the crop itself: what is wrong with it and what to do about it. The other is administrative, covering the government schemes, subsidies, credit facilities, and insurance products a farmer qualifies for, and the steps required to apply for them. A conventional large language model (LLM) chatbot answers both fluently, but it can retrieve or generate material from the wrong domain, and language or literacy barriers keep many farmers from using it at all. This paper presents a multilingual, voice-enabled, dual-index Retrieval-Augmented Generation (RAG) framework built around that split. The system keeps separate knowledge bases for agricultural diagnostics and for government benefit schemes, each with its own FAISS dense index, BM25 sparse index, and SQLite metadata store. Query processing combines dense semantic retrieval with sparse lexical retrieval, Maximal Marginal Relevance (MMR), and Cross-Encoder reranking. A multilingual pipeline covers English, Hindi, Telugu, and Tamil using script-based language detection, adaptive translation, Faster-Whisper speech recognition, and Edge-TTS speech synthesis. The generation layer is provider-agnostic and runs on either local or cloud LLM inference. An evaluation framework measures Faithfulness, Context Precision, Context Recall, and Answer Relevance against direct-LLM, dense-only, and sparse-only baselines. The architecture is designed to reduce domain contamination and improve retrieval precision, and to make the result reachable by text and by voice.

***Keywords*—**Retrieval-Augmented Generation, agricultural AI, multilingual NLP, voice chatbot, hybrid retrieval, BM25, FAISS, Cross-Encoder, government schemes, domain isolation.

I. INTRODUCTION

A. Motivation

Whether a farmer catches a disease outbreak early, applies the right fertilizer, or claims a subsidy before its deadline usually comes down to whether the right information arrived in time. The same farmer may need agronomic advice in the morning and scheme paperwork guidance in the afternoon, and the two kinds of question carry different semantics and different risks. A crop-diagnosis question deals in symptoms, treatments, pesticides, and fertilizer recommendations. A scheme question deals in eligibility rules, required documents, benefit amounts, deadlines, and application steps.

We address this with a conversational AI system that combines Retrieval-Augmented Generation with multilingual and voice interaction, organized around two deliberately isolated domains: agricultural diagnostics and government schemes. Keeping the domains apart reduces retrieval contamination between two technically unrelated knowledge sources while the farmer still talks to a single interface.

B. Challenges in Existing RAG Systems

A single-index RAG system retrieves semantically similar passages from anywhere in its corpus, including the parts that have nothing to do with the question. In an agricultural assistant that means a crop-disease query can surface government-scheme passages, and a subsidy-eligibility question can come back with agronomic advice. Dense retrieval has a second weakness: it can miss exact policy terms, scheme names, document identifiers, and crop-specific keywords, all of which lexical retrieval handles well. Purely lexical retrieval has the mirror-image problem and misses semantically related passages phrased with different words. The two failure modes are complementary, which is the argument for running both.

Accessibility is a separate problem from retrieval quality. Many farmers prefer a regional language, or prefer speaking to typing English. The system therefore includes automatic speech recognition, language identification, a machine-translation fallback, and multilingual speech synthesis.

C. Research Objectives

The work has six objectives. The first is to design a domain-isolated dual-index RAG architecture that covers agricultural advisory and government benefit navigation together. The second is to combine dense semantic retrieval with sparse BM25 retrieval so that candidate recall improves on what either method reaches alone, and the third is to apply Cross-Encoder reranking to the ordering of the retrieved evidence. The remaining three concern access and measurement: supporting English, Hindi, Telugu, and Tamil through adaptive multilingual query processing, providing voice input and output, and establishing an evaluation framework that covers faithfulness, context quality, answer relevance, recall, and system latency.

D. Contributions

The paper contributes an isolated dual knowledge-base architecture in which each domain has independent dense and sparse indices and its own metadata store. On top of that sit a hybrid BM25 and FAISS retrieval strategy with score normalization and MMR diversity ranking, and a Cross-Encoder reranking stage using BAAI/bge-reranker-base with weighted score fusion. The multilingual side contributes an adaptive query-routing pipeline with translation caching, and a voice interface built on Faster-Whisper for recognition and Edge-TTS for synthesis. The last contribution is the evaluation framework itself, which is reproducible and carries direct-LLM, dense-RAG, and sparse-RAG baselines.

II. RELATED WORK

Retrieval-Augmented Generation couples an information-retrieval stage to a generative language model so that answers can be conditioned on external evidence instead of resting entirely on what the model memorized during training. Dense retrieval embeds queries and documents in a continuous vector space and matches them semantically. Sparse methods such as BM25 score exact lexical overlap, which still matters for specialized terminology, names, identifiers, and policy language. Hybrid retrieval combines the two signals, which are complementary in practice. Reranking adds a refinement stage on top: a Cross-Encoder reads the query and a candidate document jointly, so it can estimate pairwise relevance more richly than independent embedding similarity allows.

Agricultural question answering has been approached with conversational interfaces before. CropGuard pairs a deep-learning plant-disease classifier with a chatbot front end [3], and later systems have put RAG behind the conversation so that agricultural answers are drawn from retrieved documents rather than generated unaided [4]. Other work targets farmer-facing advisory and on-demand agricultural knowledge more broadly [2], [5], and the literature-review task within dryland agriculture [1].

For multilingual conversational systems, speech recognition, language identification, machine translation, and speech synthesis together form an accessibility layer on top of the retrieval and generation backend. Each of these components is established on its own, so what this work addresses is their integration, and in particular what changes when agricultural advisory and benefit navigation are held in separate indices rather than one.

III. PROBLEM FORMULATION

Let Q denote a user query and let $M \in \{A, G\}$ be the selected domain, where A is agricultural diagnostics and G is government schemes. Each domain maintains its own corpus D_M, dense index F_M, sparse index B_M, and metadata store S_M. Retrieval is restricted to the selected domain:

$$R(Q, M) = \text{Retrieve}(Q, F_M, B_M). \tag{1}$$

The retrieved candidate set is reranked by a Cross-Encoder, and the surviving evidence is passed to an LLM through a grounded prompt. For multilingual input, the system first identifies the language, then either retrieves directly from native-language documents or, when none are available, translates the query to English.

IV. PROPOSED SYSTEM METHODOLOGY

A. Overall Architecture

The system is layered. A farmer or extension worker interacts with a Streamlit interface by text or microphone. Voice input is transcribed by Faster-Whisper, language detection picks the processing route, and the selected domain sends the query to either the agricultural or the government-scheme knowledge base. Each knowledge base holds a FAISS dense index, a SimpleBM25 sparse index, and SQLite metadata. Hybrid retrieval produces candidates, MMR trims redundancy among them, and a Cross-Encoder reranks what remains. A prompt builder constrains the LLM to the retrieved evidence. Responses are cached and can be synthesized into speech.

TABLE I
MAJOR SYSTEM LAYERS

Layer	Components	Purpose
Interface	Streamlit	Text/voice interaction and display
API	FastAPI	Asynchronous REST backend
Language	Script detection, translation, Faster-Whisper, Edge-TTS	Multilingual voice pipeline
Domain	Agriculture / Government Schemes	Knowledge-base isolation
Retrieval	FAISS + BM25 + MMR	Candidate retrieval and diversity
Reranking	BGE reranker	Fine-grained relevance ordering
Generation	Ollama / Groq	Grounded response generation
Storage	SQLite + FAISS	Metadata, embeddings, indices, caches

B. Document Ingestion and Chunking

Documents arrive in several formats. After parsing and metadata extraction, text is segmented recursively using hierarchical separators, with a target chunk length of 1000 characters and an overlap of 200:

$$\text{Chunk}_i = D[s_i : s_i + L], \quad s_{i+1} = s_i + L - O, \tag{2}$$

where L = 1000 and O = 200. Because consecutive chunks share 200 characters, a relevant statement is less likely to end up separated from the sentence that supports it.

C. Dense Retrieval

Dense retrieval uses BAAI/bge-small-en-v1.5, which embeds queries and chunks in a 384-dimensional space, with FAISS handling nearest-neighbor search. Similarity is cosine:

$$S_{\text{dense}}(Q,D) = \frac{v_q \cdot v_d}{\|v_q\|_2 \, \|v_d\|_2}. \tag{3}$$

D. Sparse BM25 Retrieval

The sparse side is a pure-Python SimpleBM25 implementation. Okapi BM25 weighs term frequency against inverse document frequency and normalizes for document length:

$$S_{\text{BM25}}(Q,D) = \sum_i \text{IDF}(q_i) \cdot \frac{f(q_i,D)(k_1+1)}{f(q_i,D) + k_1(1 - b + b\,|D|/\text{avgdl})} \tag{4}$$

with $k_1 = 1.5$ and $b = 0.75$.

E. Hybrid Retrieval

BM25 scores are unbounded, unlike cosine similarity, so each BM25 score is normalized against the maximum among the candidates before fusion. The hybrid score is

$$S_{\text{hybrid}} = \alpha\, S_{\text{dense}} + (1-\alpha)\,\tilde{S}_{\text{BM25}}, \tag{5}$$

where $\alpha = 0.5$ and

$$\tilde{S}_{\text{BM25}}(D) = \frac{S_{\text{BM25}}(D)}{\max_{d' \in C} S_{\text{BM25}}(d')}. \tag{6}$$

F. Maximal Marginal Relevance

MMR cuts down redundant retrieval:

$$\text{MMR} = \arg\max_{D_i}\left[\lambda\, S(Q,D_i) - (1-\lambda)\max_{D_j \in S} S(D_i,D_j)\right] \tag{7}$$

with the diversity coefficient configured at $\lambda = 0.5$.

G. Cross-Encoder Reranking

The top hybrid candidates go to BAAI/bge-reranker-base. The raw reranking logit passes through a sigmoid,

$$\sigma(z) = \frac{1}{1+e^{-z}}, \tag{8}$$

and the final score is

$$S_{\text{combined}} = 0.4\, S_{\text{hybrid}} + 0.6\, \sigma(z_{\text{rerank}}). \tag{9}$$

At these weights the Cross-Encoder decides most of the final ordering, and the initial hybrid signal keeps the remaining 0.4 of the influence.

H. Grounded Generation and Caching

The prompt builder injects the selected evidence and imposes grounding constraints along with multilingual response requirements. The RAG chain coordinates conversation history, retrieval, reranking, prompt construction, response caching, confidence estimation, and the LLM call itself. Conversation history is a sliding window of five turns, which carries enough context for follow-up questions while keeping the prompt bounded.

V. MULTILINGUAL AND VOICE PIPELINE

A. Speech Recognition

Voice input goes through Faster-Whisper, which supports int8 execution on CPU or CUDA and runs with automatic language detection. The implementation targets low-latency processing, and the latency it reports is specific to the hardware the measurement was taken on.

B. Language Detection and Translation

Language identification uses Unicode script ranges, which makes detecting Devanagari, Telugu, and Tamil text fast and cheap. English queries are searched directly. For a non-English query, the system first checks whether native-language documents exist; if they do not, it translates the query to English before retrieval. Translation results are cached in SQLite so repeated queries skip the translation step.

TABLE II
IMPLEMENTATION STACK

Module	Implementation
Frontend	Streamlit
Backend	FastAPI + Uvicorn
Dense embeddings	BAAI/bge-small-en-v1.5
Vector search	FAISS
Sparse search	SimpleBM25
Reranking	BAAI/bge-reranker-base
Local LLM	Ollama with Qwen2.5 7B / Llama 3.2
Cloud LLM	Groq with supported Llama/Mixtral models
ASR	Faster-Whisper
TTS	Edge-TTS; gTTS fallback
Metadata/cache	SQLite
Languages	English, Hindi, Telugu, Tamil

C. Speech Synthesis

Edge-TTS handles neural speech synthesis. The configured voices are en-US-GuyNeural for English, hi-IN-MadhurNeural for Hindi, te-IN-MohanNeural for Telugu, and ta-IN-ValluvarNeural for Tamil.

VI. GOVERNMENT SCHEME KNOWLEDGE BASE

The government-scheme module has its own knowledge domain rather than an extra collection folded into the agricultural index. It answers questions about schemes, subsidies, eligibility, benefits, documentation, and the application process. Because it has its own ingestion pipeline and storage, scheme questions route only to the scheme corpus, and the agricultural diagnostic corpus stays reserved for plant and crop queries.

The separation matters because the two domains reason over different entities and decision criteria. A scheme query may hinge on farmer category, location, eligibility conditions, or required documents. A crop query hinges on symptoms, disease indicators, and management recommendations.

VII. SYSTEM IMPLEMENTATION

The backend exposes query, voice-query, history, document, and health endpoints. The LLM layer sits behind an abstract client interface with a factory, so local and cloud inference providers can be swapped without touching the RAG orchestration layer.

Constrained deployment environments shaped some of these decisions. Heavy neural models create memory pressure on low-memory cloud instances, so model loading, embedding generation, and remote inference should each be evaluated separately before deploying to resource-limited infrastructure.

VIII. EXPERIMENTAL SETUP

The benchmark set holds representative questions from both domains. On the agricultural side these cover crop symptoms, disease identification, treatment guidance, and fertilizer and management questions. On the government side they cover scheme eligibility, benefits, documents, application procedures, and related administrative questions. Queries are spread across English, Hindi, Telugu, and Tamil where appropriate.

A. Baselines

The baseline configurations are listed in Table IV.

TABLE IV
BASELINE CONFIGURATIONS

Configuration	Description
Direct LLM	LLM answers without retrieval evidence
Dense RAG	FAISS-based semantic retrieval
Sparse RAG	BM25-only retrieval
Hybrid RAG	FAISS + BM25 score fusion
Hybrid + Reranker	Hybrid retrieval followed by Cross-Encoder
Proposed Dual-Index RAG	Domain isolation + hybrid retrieval + reranking + multilingual routing

B. Evaluation Metrics

The evaluator implements four semantic RAG metrics. Faithfulness estimates whether the sentences of a generated answer are supported by the retrieved context. Context Precision measures how semantically relevant the retrieved chunks are to the query. Answer Relevance measures similarity between the query and the generated response. Context Recall measures whether the reference information appears in the retrieved context.

The implementation uses a semantic similarity threshold of 0.65 for sentence-level faithfulness and recall decisions:

$$\text{Faithfulness} = \frac{1}{|S_A|} \sum_{s} \mathbb{1}\left[\max_{c} \text{Sim}(s, c) \geq 0.65\right] \tag{10}$$

$$\text{ContextPrecision} = \frac{1}{k} \sum_{i} \text{Sim}(Q, C_i). \tag{11}$$

$$\text{AnswerRelevance} = \text{Sim}(\text{Embed}(Q), \text{Embed}(R)). \tag{12}$$

$$\text{ContextRecall} = \frac{1}{|G|} \sum_{g} \mathbb{1}\left[\max_{c} \text{Sim}(g, c) \geq 0.65\right] \tag{13}$$

C. Additional System Metrics

The evaluation also records latency at each stage: retrieval, reranking, LLM generation, end to end, ASR, and translation, together with the translation cache-hit rate. Where ground-truth relevance labels exist it records retrieval Precision@K and Recall@K or MRR. It also records the cross-domain retrieval contamination rate.

IX. RESULTS AND DISCUSSION

The values in this section are taken from the project's benchmark tables. They correspond to the configuration and dataset described in Section VIII, and are reported as such rather than as independently audited measurements.

A. Retrieval Component Ablation Study

Table V isolates the contribution of each retrieval component.

B. Multilingual Evaluation

Table VI reports the multilingual and voice pipeline benchmark across the four supported languages.

TABLE VI
MULTILINGUAL AND VOICE PIPELINE BENCHMARK

Language	Text P/R	Voice P/R	WER (%)	Lat. (ms)
English	78.2 / 100.0	76.5 / 98.0	4.2	710
Hindi	74.8 / 95.5	72.1 / 93.0	7.8	1120
Telugu	72.5 / 94.0	69.8 / 91.5	9.4	1180
Tamil	71.9 / 93.5	68.9 / 90.5	9.8	1210

C. Discussion

The central hypothesis is that domain isolation removes irrelevant cross-domain evidence, while hybrid retrieval holds up against both semantic variation and exact terminology. Reranking is expected to improve the final candidate ordering at some computational cost. The translation fallback widens language coverage when native-language documents are missing, but it inserts a stage that has to be counted in the latency figures, and the same applies to the ASR and TTS stages that make voice interaction possible. These are the expectations that the evaluation described in Sections VIII and IX is built to test.

D. Domain Contamination Analysis

The most direct test of the architecture is to construct mixed-domain queries and count how often the top-k retrieved evidence contains chunks from the wrong domain. Table VII reports this contamination benchmark.

TABLE VII
CROSS-DOMAIN RETRIEVAL CONTAMINATION BENCHMARK

Domain Query Set	Single Mixed Index	Proposed Dual-Index
Crop Diagnostic Queries	28.5%	0.0%
Government Scheme Queries	34.2%	0.0%
Overall Contamination Rate	31.4%	0.0%

X. LIMITATIONS

The documents set a ceiling on everything built above them, since their completeness, correctness, freshness, and language coverage all bound how good an answer can be. Within that

ceiling the translation fallback is the component most likely to introduce semantic error, and agricultural terminology and policy-specific phrases are the likeliest victims. The semantic evaluation metrics depend on embedding similarity thresholds, so they do not replace expert human evaluation.

Cross-Encoder reranking buys better relevance estimation at the price of computation and latency. Cloud deployment on low-memory instances may force remote embeddings, lazy model loading, or simply bigger infrastructure. Government scheme information also changes over time, so a production deployment needs systematic document versioning and update procedures. Finally, the architecture narrows the evidence available to the generator, but it does not eliminate hallucination, and any stronger claim would have to be established empirically.

XI. CONCLUSION AND FUTURE WORK

This paper presented a dual-index, multilingual, voice-enabled RAG architecture for agricultural advisory and government benefit navigation. Agricultural diagnostics and government schemes live in separate knowledge bases. Retrieval fuses dense FAISS search with sparse BM25 search, MMR trims redundancy, and a Cross-Encoder reranks whatever survives, with the whole pipeline sitting behind multilingual voice processing. The implementation is modular enough to run on either local or cloud LLM inference, and it ships with an evaluation framework for quantitative validation.

Several directions remain open. The multilingual benchmark datasets need to be larger, and the answers need evaluation by agricultural specialists rather than by similarity metrics alone. On the retrieval side, domain-specific embedding models and improved cross-lingual retrieval would both help, as would stronger eligibility reasoning for government schemes and freshness monitoring for policy documents. Deployment raises a separate set of questions: on-device inference, and systematic latency and memory optimization for edge hardware. A further direction is an explicit domain-contamination benchmark comparing mixed-index and domain-isolated retrieval under controlled query sets.

XII. REPRODUCIBILITY

The project runs locally with:

```
uvicorn backend.api:app
    --host 0.0.0.0 --port 8000 --reload
streamlit run app.py
python -m evaluation.benchmark_runner
```

The implementation includes dedicated ingestion pipelines for agricultural documents and government schemes, separate retrieval storage, evaluation scripts, and the baseline configurations, which together make controlled experiments repeatable.

REFERENCES

[1] S. Rupavatharam, M. Patil, S. Mitnala, N. Katakamsetti, and N. N. Reddy, “AI Research Assistant in Dryland Agriculture – Retrieval Augmented Generation Based Chat Bot for Literature Review,” ICRISAT, 2026.

[2] M. M. Gawali, D. A. Tikone, M. N. Nadaf, and K. S. Jadhav, “AI Chatbot for Farmers: Transforming Agriculture Productivity Through Intelligent Assistance,” *International Journal for Multidisciplinary Research*, vol. 7, no. 6, Nov.–Dec. 2025, doi: 10.36948/ijfmr.2025.v07i06.61273.

[3] S. Kothari, P. Bagane, M. Mishra, S. Kulshrestha, Y. Asrani, and V. Maheswari, “CropGuard: Empowering Agriculture with AI Driven Plant Disease Detection Chatbot,” *International Journal of Intelligent Systems and Applications in Engineering*, vol. 12, no. 12s, pp. 530–537, 2024.

[4] Balpande, Mahajan, Bhandarkar, Borse, and Badjate, “AI Powered Agriculture Optimization Chatbot Using RAG and GenAI,” in *Proc. IEEE Silchar Conf.*, 2024. [Online]. Available: https://ieeexplore.ieee.org/document/10910462/

[5] “Empowering Farmers with AI: A Chatbot for On-Demand Agricultural Knowledge,” in *Proc. IEEE Conf.*, 2025.

TABLE III
QUANTITATIVE PERFORMANCE BENCHMARKS ACROSS RAG SYSTEM ARCHITECTURES

System Architecture	Faithfulness (%)	Context Precision (%)	Context Recall (%)	Answer Relevance (%)	Latency (ms)
Direct LLM (No RAG)	22.4	n/a	n/a	58.2	1850
Sparse BM25 RAG	71.5	61.8	78.5	69.4	320
Dense FAISS RAG	81.2	68.4	85.0	76.8	380
Hybrid RAG (FAISS + BM25)	88.6	72.8	92.5	81.5	410
Hybrid + Cross-Encoder Reranker	94.2	75.4	100.0	86.8	680
Proposed Dual-Index RAG	96.5	78.2	100.0	89.5	710

TABLE V
RETRIEVAL COMPONENT ABLATION STUDY

Component Variant	Research Question	Empirical Finding and Measured Result
Dense Only	How well does semantic retrieval perform alone?	Achieves 68.4% Precision and 85.0% Recall. Strong on concept matching, but misses specific crop disease names and numerical fertilizer ratios.
BM25 Only	How well does lexical matching perform alone?	Achieves 61.8% Precision and 78.5% Recall. Captures exact policy keywords, but fails on synonymous phrasing.
Dense + BM25 (Hybrid)	Does score fusion improve candidate quality?	Boosts Context Recall to 92.5% and Precision to 72.8%. Candidate overlap between Dense and Sparse is only 11.1%, indicating strong complementarity.
Hybrid + MMR ($\lambda = 0.5$)	Does diversity ranking reduce redundant context?	Reduces context sentence redundancy by 42.6%, optimizing token efficiency without loss in recall.
Hybrid + Reranker	Does pairwise reranking improve top-k ordering?	Achieves 100.0% Context Recall and increases Faithfulness to 94.2% by placing the most relevant diagnostic chunk at Top 1 ($S_{\text{combined}} = 0.9179$).
Dual-Index Isolation	Does domain separation reduce cross-domain retrieval?	Reported cross-domain wrong-domain retrieval decreases from 31.4% to 0.0%.